

1(a)(i)	work (done) / time (taken) or energy (transferred) / time (taken)	B1
1(a)(ii)	Correct substitution of base units of all quantities into any correct equation for power. Examples: $(P = E / t \text{ or } W / t \text{ gives}) \text{ kg m}^2 \text{ s}^{-2} / \text{s} = \text{kg m}^2 \text{ s}^{-3}$ $(P = Fs / t \text{ or } mgh / t \text{ gives}) \text{ kg m s}^{-2} \text{ m} / \text{s} = \text{kg m}^2 \text{ s}^{-3}$ $(P = \frac{1}{2}mv^2 / t \text{ gives}) \text{ kg (m s}^{-1})^2 / \text{s} = \text{kg m}^2 \text{ s}^{-3}$ $(P = Fv \text{ gives}) \text{ kg m s}^{-2} \text{ m s}^{-1} = \text{kg m}^2 \text{ s}^{-3}$ $(P = VI \text{ gives}) \text{ kg m}^2 \text{ s}^{-2} \text{ A}^{-1} \text{ s}^{-1} \text{ A} = \text{kg m}^2 \text{ s}^{-3}$	A1
1(b)(i)	units of A : m^2 and units of T : K	C1
	units of k : $\text{kg m}^2 \text{ s}^{-3} / \text{m}^2 \text{ K}^4$ $= \text{kg s}^{-3} \text{ K}^{-4}$	A1
1(b)(ii)	curve from the origin with increasing gradient	B1

Question	Answer	Marks
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